

SIFRA-AI: Oil & Gas Workplace Safety Knowledge Base

Educational knowledge base for testing a RAG pipeline. This is not a replacement for official regulations, site-specific procedures, or emergency plans.

1. General Safety Principles

- Identify hazards before work begins, including equipment, chemicals, energy sources, work conditions, and environmental factors.
- Use the hierarchy of controls where possible: elimination, substitution, engineering controls, administrative controls, and appropriate PPE.
- Stop work and report unsafe conditions according to approved site procedures.

2. Gas Leakage and Hazardous Atmospheres

- Treat a suspected gas release or hazardous atmosphere as a serious condition and follow the approved alarm and emergency procedures.
- Do not enter potentially hazardous areas unless authorized and required testing or permit procedures have been completed.
- Control ignition sources where flammable materials may be present and follow site isolation procedures.

3. Respiratory Hazards

- Respiratory hazards may include gases, vapors, dusts, fumes, and oxygen-deficient atmospheres.
- Use approved engineering controls, ventilation, monitoring, and respiratory protection according to site procedures.
- Report suspected exposure promptly through the workplace reporting process.

4. Chemical and Poisoning Hazards

- Review labels, safety data sheets, and approved procedures before handling hazardous chemicals.
- Use approved storage, containment, transfer, PPE, and spill-response procedures.
- Do not improvise cleanup or disposal of hazardous substances.

5. Fire and Explosion Prevention

- Control ignition sources and follow permit-to-work requirements for hot work and other hazardous activities.
- Use equipment approved for the relevant work environment and follow emergency alarm procedures.

6. Equipment and Mechanical Safety

- Inspect machinery before use and do not bypass guards or safety devices.
- Follow approved energy-isolation or lockout/tagout procedures before maintenance where required.
- Only trained and authorized personnel should operate specialized equipment.

7. Incident Reporting

- Record factual details such as the task, location, equipment involved, observed hazards, timing, and immediate actions.
- Focus investigations on contributing factors and corrective actions rather than blame.
- Use lessons learned to improve controls, training, equipment, and procedures.

8. Risk Assessment

- Evaluate likelihood and severity using the organization's approved risk method.
- Escalate high-risk conditions and reassess risk when the work scope or conditions change.

9. Emergency Response

- Know the site's alarm signals, assembly points, communication methods, and emergency contacts.
- Follow the approved emergency plan and instructions from trained responders.

10. Example RAG Queries

- What should workers do after a suspected gas leak?
- What controls help reduce respiratory hazards?
- What information should be included in an incident report?
- How should equipment be made safe before maintenance?
- What factors should be considered in a risk assessment?

Suggested Use in SIFRA-AI

PDF → Text Extraction → Chunking → Embeddings → FAISS Vector Database → User Query → Retrieve Relevant Chunks → LLM → Final Safety Response.

For a production system, add official, current safety documents and retain source metadata such as document title, version, publication date, and page number.